

AI-Driven Gamified Intervention Models for Autism Education Across Developmental Stages

Shunlei Xu
School of Information Engineering
Huzhou University
Huzhou, Zhejiang, China
13868496930@163.com

Jingwen Su
UcardStore
Shanghai, China
jingwen.su@ucardstore.com

Jing Chen
The Third People's Hospital of Huzhou
Huzhou, Zhejiang, China
2787174619@qq.com

Xianwei Lin
Institute of Education and Innovation in Paris (IEIP)
Paris, Ile de France, France
heiansenlin@126.com

Zefeng Wang*
School of Information Engineering
Huzhou University
Huzhou, Zhejiang, China
zefeng.wang@zjhu.edu.cn

Abstract

This article studies the application of artificial intelligence-assisted gamification therapy in children with autism spectrum disorder of different ages. The autism spectrum disorder mentioned here is what we commonly refer to as ASD children. This research combines gamification and artificial intelligence technology to provide children with personalized learning materials to improve their skills in language, social interaction and emotion. These projects will offer personalized advice to children with autism, plan learning paths suitable for them, and provide timely feedback to help them solve specific problems they encounter. This approach also designs courses based on different development stages and creates some sensory stimulating activities for children who haven't started to speak yet. Design tools that can help vocabulary development for children who are just beginning to learn languages, and design communication exercises for children with stronger learning abilities. The results of the experiment show that the children have made remarkable improvements in terms of participation, language skills, and social emotional behaviors. Artificial intelligence has potential in supporting children with autism, and it can also be seen that artificial intelligence methods play a crucial role in autism education.

CCS Concepts

• **Applied computing**; • **Education**; • **Interactive learning environments**;

Keywords

Autism, AI-driven education, gamification, adaptive learning, social-emotional development

*Corresponding author.

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MIDA 2025, Ningbo, China
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ACM ISBN 979-8-4007-1386-6/2025/04
https://doi.org/10.1145/3744464.3744497

ACM Reference Format:
Shunlei Xu, Jingwen Su, Jing Chen, Xianwei Lin, and Zefeng Wang. 2025. AI-Driven Gamified Intervention Models for Autism Education Across Developmental Stages. In *2nd International Conference on Machine Intelligence and Digital Applications (MIDA 2025), April 18–20, 2025, Ningbo, China.* ACM, New York, NY, USA, 8 pages. https://doi.org/10.1145/3744464.3744497

1 Introduction

Artificial intelligence, or AI for short, has become a particularly powerful tool in the healthcare field. It offers innovative solutions for a wide variety of medical conditions, including mental health disorders such as childhood depression. Ai-assisted treatment has received significant attention because it has the potential to provide personalized and adaptive interventions. These interventions can meet the different needs of each patient. With the help of advanced algorithms and the capabilities of machine learning, artificial intelligence can provide personalized and adaptive solutions to meet the unique needs of each individual. This is particularly crucial in the case of children with special needs, as the traditional approach of treating all children in one way often fails to meet the requirements. Autism spectrum disorder is a complex neurodevelopmental disorder characterized by obstacles in social communication and interaction, as well as some repetitive and limited behaviors [1]. When exploring the possibility of artificial intelligence addressing the special needs of children with autism, this study found that autism spectrum disorder requires specially designed educational intervention measures to meet the needs of children at different developmental stages [2]. Although traditional methods, such as applied behavior analysis and treatment and educational approaches for children with autism and related communication disorders, It has been proved to be effective, but these methods have been criticized for being inflexible enough to adapt well to the specific needs and preferences of each person [3]. However, by introducing an adaptive feedback mechanism, artificial intelligence can adjust the teaching content according to the specific circumstances of students to meet their needs [4]. Artificial intelligence can analyze a large amount of data to identify the patterns within it, and then customize intervention measures based on these to ensure that each child can receive the most effective support. In a study called "Exploring the Impact of Gamified Co-design on the Learning and Participation

of Children with Autism in the Context of Literacy", the combination with artificial intelligence has also been proven to be an effective educational approach [5]. Gamification educational tools can significantly enhance the emotional understanding ability and social skills of children with autism [6]. In a project of designing social and emotional games for children with autism, the research shows that gamification can increase the interest and participation of children with autism in learning by combining game elements such as rewards, challenges and interactions [7]. Compared with traditional teaching methods, gamification educational tools can significantly improve the emotional understanding ability, social skills, and the quality of task completion of children with autism [8]. In addition, adding augmented reality and virtual reality technologies to the gamification platform can create immersive scenes that are very similar to the real world. Provide children with a safe environment to practice social and emotional skills [9].

2 Research foundation

Artificial intelligence-driven technologies will redefine the future direction of healthcare and can significantly improve patients' health conditions and achieve unprecedented progress. These computer systems can learn, perform and complete those tasks that generally require human intelligence to accomplish. It encompasses a series of technologies, including autonomous systems, machine learning, computer vision, and Natural Language Processing (NLP). Among the group of children with autism spectrum disorder (ASD), artificial intelligence-assisted technologies have shown promising results in improving adaptive functions [10]. The most common type of technology is robotics. Besides, and applications based on mobile phones or computers, virtual reality systems, tablet devices or screens, and wearable devices.

For children with autism, artificial intelligence-assisted treatment can offer adaptive and personalized intervention measures, thus meeting the needs of each individual [11]. Traditional educational tools often fail to adapt to the changes of children, providing the same content regardless of whether they have made progress or not. Artificial intelligence can make up for this deficiency by creating systems. This system can adjust the assistance and strategies according to the child's progress. Artificial intelligence-based applications on mobile phones or computers can provide personalized learning experiences for children with autism, adapting to their learning speed and style. Using artificial intelligence in this way can help solve the unique challenges faced by children with autism and provide them with the support they need to thrive. However, these new ideas are still relatively new at present and more work needs to be done to make them more perfect [12].

3 Methodology

3.1 Model Development

Combining intelligent computer programs with game-based learning can be regarded as a relatively significant change in our education of children with autism [13]. A crucial part of this brand-new approach is that the curriculum can be adjusted according to the growth and learning situation of the children. These intelligent computer programs seem to be the core driving force of this new teaching method. They can observe children's performance and behavior, and then make changes to the learning space on the spot. With the help of computer models that can learn from the data, these systems can understand how children answer questions. In this way, they can arrange courses for each child that are suitable for their current abilities and the content they need to learn next. By providing timely feedback and gradually increasing the complexity of tasks, artificial intelligence can support a progressive learning process that is in line with children's natural development. Integrating natural language processing technology into AI-driven models, by recognizing speech patterns and providing context-related corrections, it can promote children's transformation from non-verbal communication to verbal communication. Combining artificial intelligence and game-based learning into a unified system is supported by the information we obtain from the data. As time goes by, this can help make the system more and more perfect (as shown in Figure 1). By viewing a large amount of information from many users, these models will become smarter in infering the needs of each learner and providing the most appropriate content. This continuous feedback loop can ensure that this system keeps evolving and changing along with new discoveries in teaching techniques and autism research, and it can provide solutions for an increasing number of learners.

3.2 Data Collection and Analysis

3.2.1 Data Sources. Effective data collection and analysis are crucial for understanding the educational needs of children with autism spectrum disorder (ASD) [14]. In this study, data sources are divided into two primary categories: behavioral data and feedback from caregivers and educators (this data was approved by the Medical Ethics Committee of Huzhou Third People's Hospital). These sources provide complementary insights, allowing for a comprehensive understanding of intervention effectiveness.

Behavioral data is a key aspect when evaluating educational intervention for ASD. Indicators such as interaction frequency and task completion rate are mostly used to assess children's participation in gamified learning tools and educational content. Behavioral data can generally capture those subtle but crucial aspects in the learning process, such as sustained attention. And the ability to switch between different activities. Because of the recent advancements in digital tracking technologies such as GeniAuti and EEG, caregivers can now systematically monitor children's problem behaviors and interactions(as shown in Figure 2). To provide targeted intervention measures, these systems will offer real-time data on behavioral trends [15]. To meet the needs of each individual, The development of educational resources is guided by behavioral indicators obtained from interactive data, which can provide some key information about the participation level of students with autism.

The collection of data for autism education requires the efforts of educators and caregivers. Caregivers can offer unique perspectives on the family environment and explain in detail how children apply the skills they have learned in daily life. Teachers, on the other hand, observe how students interact in the classroom and how their peers participate. And how they accomplished the task. Studies have shown that including nursing staff in applied behavior analysis research can improve the quality of the collected data. After they receive systematic behavior recording training, the digital system

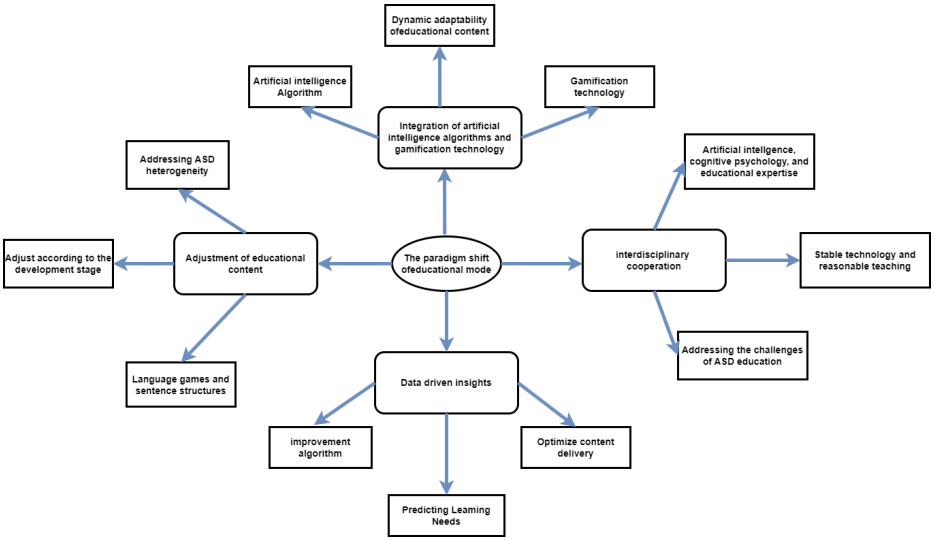


Figure 1: The paradigm shift of education mode

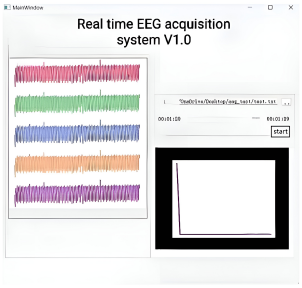


Figure 2: Children's EEG signals collected using devices

can effectively integrate the input reported by nursing staff, so that learning problems and milestones can be monitored regularly. Such information is helpful for confirming whether the treatment is effective. And it is very necessary to make the treatment suitable for specific needs.

3.2.2 Analysis Methods. When looking at data in research on teaching kids with autism, we need to take a wide view. This means that the progress of learning needs to be digitized, as well as getting a feel for how kids are doing socially and emotionally. Using both of these ways together helps us look at the help we're giving in a full way.

Quantitative Analysis of Learning Progress

Quantitative analysis is important for evaluating the effectiveness of educational interventions for autism. By collecting data such as task accuracy, completion and the number of activities children participate in, educators can understand the learning effect. Common statistical methods, such as repeated measure Analysis of Variance (RM-ANOVA) or Hierarchical linear models (HLM), can help observe changes over time. In addition, tools such as heat maps can visually show patterns of interaction, allowing teachers to see trends in children's participation in activities, thereby identifying where more attention and support is needed. The quantitative analysis also includes a progress tracking system designed specifically for autism education. For example, Personalized Education Plans (IEPs) use statistical tools to monitor student achievement and ensure that educational programs are tailored to students' needs. Using technologies such as wearables or artificial intelligence platforms (as shown in Figure 3), researchers can adapt interventions in the first place, making them more precise and realistic.

Qualitative Evaluation of Social and Emotional Engagement

Although quantitative methods can measure functional outcomes, understanding children's sensory and social activities is also important in autism education. To do this, researchers gather information through conversations, group discussions, and observations with caregivers by finding common ground or building basic theories. Doing so can help them understand the impact of educational measures on children's emotional and social abilities.

4 AI-Driven Gamified Education Models

Games for educational purposes based on AI technology are actually new and innovative and are significantly altering the way we learn by making it more fun, a lot more personalized and also allowing a much greater level of flexibility in the process. These games have been made in a way that has made it possible to adapt them to the requirements of each individual learner which is most obvious with children suffering from Autism who need to follow special learning methods. Through techniques such as creating custom learning paths, personal recommendations, and quick feedback, these games act as a full educational solution.



Figure 3: Simultaneously record EEG signals while children are playing games, with the EEG signals located in the lower right corner of the computer screen

4.1 Adaptive Learning Paths

AI-based game learning models with artificial intelligence at its core can never function without adaptive learning lines (paths). These systems modify activities in a way of a learning a learner might progress, enabling a tailor-made educational experience to be created. Adaptive algorithms mainly cater to children with ASD who usually possess varying learning rates and styles by giving specific support. When a student finds an activity very difficult, the system is able to make it easier or provide extra help in order to avoid frustrations. However, if the learner has already mastered the task, then the platform will give more complex tasks to keep them growing and wanting to learn. This flexibility helps build academic skills, as well as boosts confidence and determination.

4.2 Personalized Recommendations

Personalized content recommendation is another fundamental function of AI-driven gamification systems. These suggestions are derived through data analysis, which assesses the interaction history of learners, their preferences, and their performance patterns. For children with autism, this function enables educators to design learning experiences that align with children's interests, thereby enhancing children's motivation and participation in learning.

4.3 Real-Time Feedback

Real-time feedback mechanism is crucial for keeping learners engaged in gamified education. Immediate feedback can help children with autism understand the consequences of their own behaviors and also help them understand causal relationships that are generally not easy to master. Multi-sensory feedback that combines visual, auditory and tactile signals can enhance such an impact. Just like the visual effects of fireworks, the auditory rewards like applause, and the vibrating touch of tablet computers, all these can create a positive experience and promote the development of good behaviors when students successfully complete tasks.

4.4 Stage-Specific Task

Gamified education based on artificial intelligence provides a flexible and adaptable approach for children with autism spectrum disorder (ASD) to meet their specific educational needs. These methods are particularly useful when adapting to the development stages of children, as they enable learners to participate in tasks suitable for their age and ability through interactive technologies, and create a structured and flexible learning environment by conducting targeted teaching in the non-verbal stage, early vocabulary stage and advanced language stage. The AI-driven gamified education model.

5 Case Studies and Experimental Results

5.1 Background of Case Studies

The case under study is to be the part of the project seeks to assess the effectiveness which will focus on the AI-driven gamification interventions for the children , who having ASD during various developmental stages and a determination of the ASD in children shall be made. The primary goal is to evaluate how customized educational tools can enhance language, social, and emotional outcomes, as well as to employ machine learning to classify children with ASD based on EEG data collected during play. Participants

were carefully chosen to represent a range of developmental challenges and communication needs, ensuring a thorough evaluation of the intervention measures.

The study included 30 children diagnosed with ASD, all aged between 4 and 12 years old (this data was approved by the Medical Ethics Committee of Huzhou Third People's Hospital). They were divided into three developmental stages: pre-verbal, early vocabulary, and advanced language. Each child was given an initial test to understand their initial language abilities, level of social engagement and emotional responses. These test results helped design personalized interventions that were incorporated into AI-powered gamification platforms.

Intervention Settings not only have a controlled environment, such as a treatment room, but also a natural environment, such as a home, where skills can be observed in different situations. Each participant is helped by a team of speech therapists, special education teachers, and caregivers who ensure their growth is multifaceted. The game tools used during the intervention are designed to meet the specific needs of each individual, using artificial intelligence algorithms to automatically adjust the difficulty and content of the tasks based on real-time performance data from the participants.

By carefully understanding the background, setting, and purpose of the participants, this section sets the stage for the subsequent analysis of the experimental design, results, and impact. Concurrently, the study also collected EEG data from children with ASD and typically developing peers. Among various machine learning methods, such as ada-boost algorithm, GaussianNB algorithm, Gradient Boosting algorithm, knn, Linear Regression method, Logistic Regression method, Random Forest, etc., Deep Neural Networks (DNN) demonstrated the highest accuracy in classifying collected individual EEG data and distinguishing between typically developing children and those with ASD (as shown Figure 4, Figure 5, Figure 6). This structured approach ensures that interventions can address the unique needs of each developmental stage, providing valuable insights into their effectiveness and scalability.

5.2 Experiment Design and Data Collection

The research conducted in this article was carefully designed to evaluate the impact of AI-driven educational games on children with autism. To ensure the accuracy and validity of the research results, this article adopted the method of digital data as well as the method of observational data. Adopting these two methods can help us understand what kind of impact these intervention measures will have at three different development stages, namely the pre-speech stage, the early vocabulary stage and the advanced language stage.

5.2.1 Experimental Design. In the experiment, the children used the gamification platform for 12 weeks. Train for 30 minutes five times a week. The platform uses artificial intelligence to adjust the difficulty of tasks according to the child's performance, ensuring that the learning experience is personalized and appropriately challenged.

5.2.2 Data Collection. During the entire intervention process, this paper collected numerical information and descriptive information, and monitored these key measurement values. :

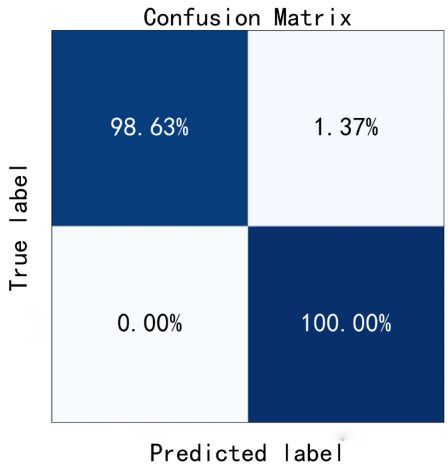


Figure 4: Confusion matrix after machine learning classification

Active Involvement Rate: This paper recorded the regularity of the participation activities and the duration of the participation activities, doing so to evaluate the system's ability to maintain concentration and curiosity. Artificial intelligence operations have a log recording mechanism, which records the length of the session, the completion rate of tasks, and the frequency of exits (As shown in Figure 7, Figure 8).

Linguistic Growth: Observations of pre-verbal children included assessing gesture usage and attempts to communicate non-verbally. In the case of initial vocabulary learners, we monitored word identification and utilization, whereas advanced learners assessed sentence complexity and conversational alternating between speaking and speaking.

This article will focus on the eye contact of people in the activities, the collective attention status and emotional responses. The feedback given by nursing staff and the notes recorded by therapists can provide more insights into the evolving nature of social interactions beyond the meeting environment.

The qualitative data collection work carried out in this article includes video recording of intervention meetings, conducting structured interviews with nursing staff, and collecting reports from therapists. These observations can capture some subtle behaviors, such as the spontaneous use of language or a high willingness to participate in group activities. Integrating quantitative indicators and qualitative insights will generate a comprehensive dataset, which can be used to properly evaluate the effects of intervention measures.

5.2.3 Ethical Considerations. This study adhered to ethical guidelines for research involving children with disabilities. Informed consent was obtained from parents or legal guardians, and where appropriate, consent was also sought from the participants (as shown in Figure 9). Additionally, the collected data was approved by the ethics committee. Gratitude is extended for the support

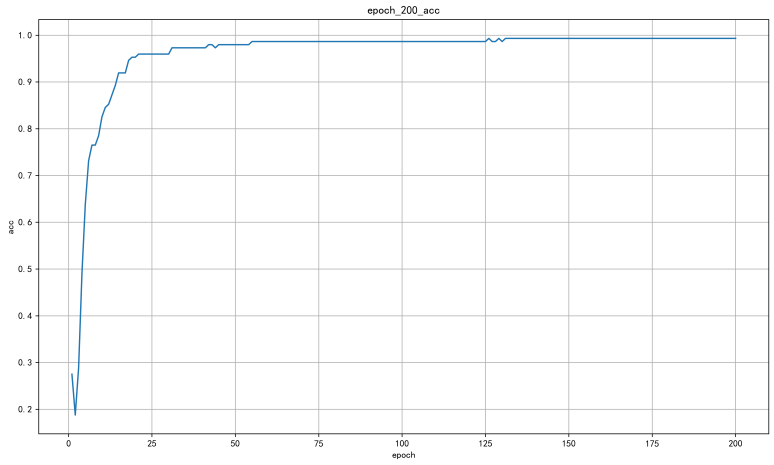


Figure 5: Line graph of accuracy after machine learning classification

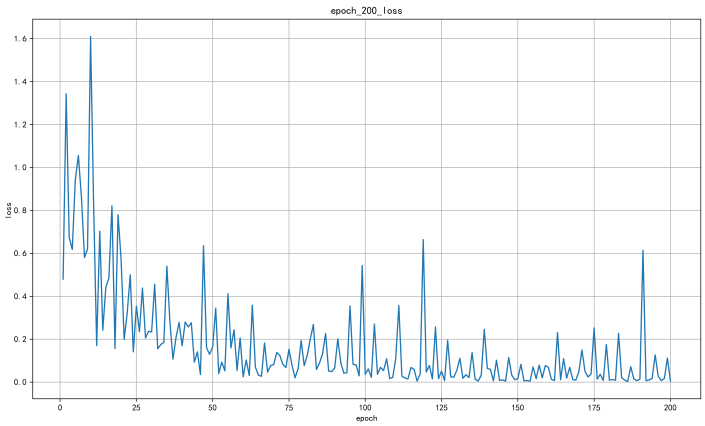


Figure 6: Line graph of loss function after machine learning classification

provided by the ethics committee. Measures were taken to ensure the well-being of participants.

By integrating a rigorous experimental design with meticulous data collection methods, a thorough evaluation was performed by professional doctors post-experiment on both the experimental group receiving AI-driven intervention and the control group undergoing traditional treatment methods. The insights derived from this approach laid the groundwork for analyzing the effectiveness, challenges, and potential of these tools in improving the education of children with ASD.

5.3 Results and Discussion

This paper assesses the results of AI-driven gamification intervention in three development stages: pre-speech, early vocabulary, and advanced language. The assessment is conducted using quantitative indicators and qualitative observations. The results obtained from the study emphasize the role of this intervention in enhancing language skills, social skills, and emotional skills, while also identifying some challenges. These challenges can provide some information for future improvements.

5.3.1 Quantitative Results. Comparing pre and post-treatment, participants demonstrated significant enhancements in engagement, language development, and social behavior. We figured out how much they paid attention by seeing how often they finished tasks and how long they stuck with a session. After the treatment, on average, they paid 15% more attention (with a standard deviation of 2.45). This shows that turning learning into a game really helps



Figure 7: Design of Child Friendly EEG Headwear Device

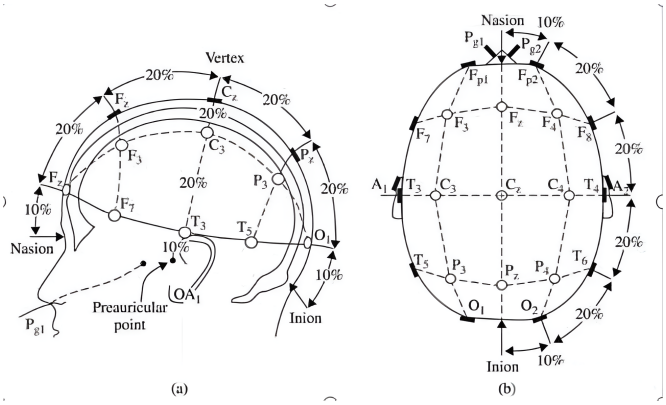


Figure 8: Schematic diagram of electrode layout for EEG acquisition equipment



Figure 9: Professional personnel assist autistic children in conducting EEG activity measurements

keep people focused and interested.In terms of language development, children in the pre-verbal stage showed an average increase of 18% (with a standard deviation of 3. 16) in non-verbal communication attempts, such as gestures and eye contact. Early vocabulary learners saw an average improvement of 24% (with a standard deviation of 3.6) in word recognition and usage accuracy. Advanced learners improved by 17% (with a standard deviation of 2.65) in sentence complexity and conversational turn-taking. These numbers tell us that making learning fit each person and changing as they learn really helps them get better with language. Social and emotional metrics also indicated marked improvements. Participants displayed higher levels of joint attention, more frequent positive emotional responses, and reduced signs of frustration during sessions (as shown in Figure 10).

5.3.2 Qualitative Observations. Input from caregivers and therapists really highlighted how much the help we gave changed how the kids acted and felt about learning. Caregivers saw that the kids were more open to joining in with groups, showed their feelings more, and didn't pull away as much when they were with others. Therapists noticed the kids got much better at feeling confident in real social situations. Kids kept going more and got less worried when they had tough tasks. Therapists think this is because the game-like learning gave them feedback right away and used lots of their senses.

6 Conclusion

This study shows how much game-like learning with AI can help in teaching kids with autism. This research also mixes ideas about teaching, how the mind works, and AI to make a guide for future stuff. Feedback in the moment and learning paths that change with the kid help them stay interested and keep trying. In society, these methods help kids fit in socially and be more on their own by helping them get past talking and communication problems. Having caregivers and teachers involved makes learning stronger in all kinds of places. Studies that follow kids over a long time and new tech like brain-computer links and virtual reality can make learning even better. It's super important for teachers, mind experts, AI makers, and rule creators to work together to break down barriers and help everyone be included. In summary, this research propels autism education forward through AI and gamification, empowering ASD learners and advocating for an inclusive educational environment.

Acknowledgments

This research was funded by the Zhejiang Provincial Education Science Planning General Project (No. 2022SCG036), titled 'Study on New Intervention Education Methods for Children with Autism Based on AI-Guided Games'. We sincerely thank the Department of Pediatric Rehabilitation, The Third People's Hospital of Huzhou in China, for their support in this study. Their assistance and collaboration were invaluable to the completion of this research.

References

[1] G. A. Abayeva, A. I. Akhmetzyanova, L. A. Butabayeva, S. K. Abildina, and A. N. Umirbekova. 2024. The ways of communication for children with autism spectrum disorder. Online Journal of Communication and Media Technologies, 14(4), e202462. https://doi.org/10.30935/ojcmt/15611.

[2] O. P. Adako, O. C. Adeusi, and P. A. Alaba. 2024. Revolutionizing Autism Education: Harnessing AI for Tailored Skill Development in Social, Emotional, and Independent Learning Domains. Journal of Computational and Cognitive Engineering 3, 4 (2024), 348-359.

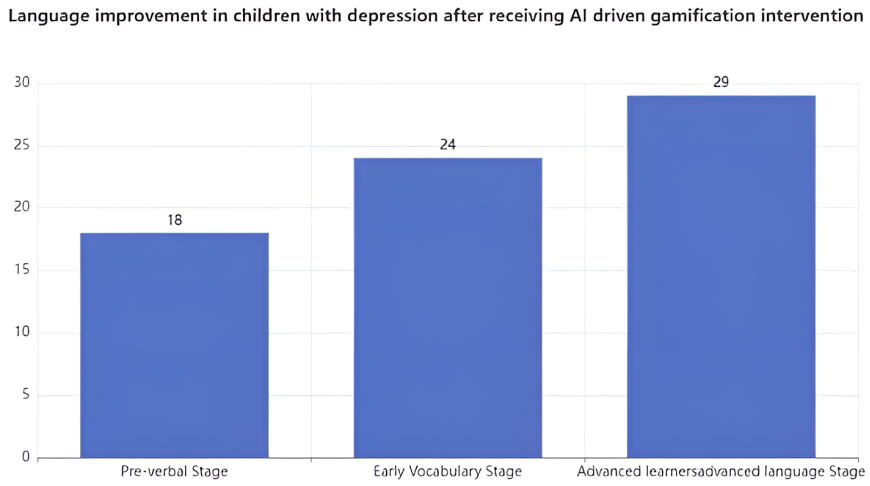


Figure 10: Language improvement in children with depression after receiving AI driven gamification intervention

[3] L. R. Silva and A. M. Toda. 2024. Effects of a Collaborative Gamification on Learning and Engagement of Children with Autism. Universal Access in the Information Society. https://doi.org/10.1007/s10209-024-01119-w.

[4] S. Mishra. 2024. Revolutionizing Education: The Impact of AI-Enhanced Teaching Strategies. International Journal For Science Technology And Engineering 12, 9 (2024), 9–32. https://doi.org/10.22214/ijraset.2024.64127.

[5] N. Xing. 2024. Artificial Intelligence to support Children with Autism. Journal of AI-Powered Medical Innovations 2, 1 (2024), 31–43. https://doi.org/10.60087/vol2iisue1.p43.

[6] L. Y. Tseng, H. C. Lee, N. Y. Chiu, *et al.* 2024. Characterizing Autism Spectrum Disorder through Fusion of Local Cortical Activation and Global Functional Connectivity using Game-Based Stimuli and a Mobile EEG System. IEEE Transactions on Neural Systems and Rehabilitation Engineering: A Publication of the IEEE Engineering in Medicine and Biology Society (2024), PP.

[7] Y. Lyu, P. An, H. Zhang, *et al.* 2024. Designing AI-Enabled Games to Support Social-Emotional Learning for Children with Autism Spectrum Disorders. arXiv preprint arXiv 2404.15576 (2024).David Harel. 1979. First-Order

[8] A. W. Moustafa and S. Ahmed. 2021. Toy Solutions for Children with Autism Spectrum Disorder Using Gamification. Autism 26, 2 (2021), 455-467. https://doi.org/10.1177/13623613211065545.R Core Team. 2019. R: A Lang.

[9] A. S, S. Seema, and H. S. Hotagi. 2023. Guiding Special Need Children using AR/VR Technology. In Proceedings of the Conference on Sustainable Innovation and Technology for Society 60515. IEEE, 1–6. https://doi.org/10.1109/csitss60515.2023.10334086.John R. Smith and Shih-Fu Ch.

[10] R. Sumra, and B. Sumra. 2024. Technology-Based Interventions for Autism Spectrum Disorder: A Systematic Review. Indian Scientific Journal of Research In Engineering And Management. 08, 11 (Nov. 2024), 1–4. DOI:https://doi.org/10.55041/ijsrem7644.

[11] O.P. Adako, O.C. Adeusi, and P.A. Alaba. 2024. Revolutionizing Autism Education: Harnessing AI for Tailored Skill Development in Social, Emotional, and Independent Learning Domains. Journal of Computational and Cognitive Engineering. (Aug. 2024). DOI:https://doi.org/10.47852/bonviewjcce42023414.

[12] Q. Yang, H. Lu, D. Liang, S. Gong, and H. Feng. 2024. Surprising Performances of Students with Autism in Classroom with NAO Robot. arXiv preprint arXiv 2407.12014 (2024). https://doi.org/10.48550/arXiv.2407.12014.

[13] I. Chaidi and A. Drigas. 2023. Digital Gaming and Autistic Spectrum Disorder. International Journal of Emerging Technologies in Learning 18, 22 (2023), 4–23. https://doi.org/10.3991/ijet.v18i22.34497.

[14] A. Aldrees, S. Ojo, J. A. Wanliss, M. Umer, M. A. Khan, B. Alabdullah, S. Alsubai, and N. Innab. 2024. Data-centric automated approach to predict autism spectrum disorder based on selective features and explainable artificial intelligence. Frontiers in Computational Neuroscience 18 (2024). https://doi.org/10.3389/fncom.2024.1489463.

[15] E. Jo, S. Park, H. Bang, Y. Hong, Y. Kim, and J. Choi. 2022. GeniAuti: Toward Data-Driven Interventions to Challenging Behaviors of Autistic Children through Caregivers' Tracking. ACM Transactions on Interactive Intelligent Systems 12, 2 (2022), 15. https://doi.org/10.1145/3512939.